



Instituto de Tecnologia e Liderança

## **Public Report**

Intelligent Platform for Social Engagement

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## Summary

|                                                            |           |
|------------------------------------------------------------|-----------|
| <b>1. Document Version Control.....</b>                    | <b>3</b>  |
| <b>2. Introduction.....</b>                                | <b>3</b>  |
| <b>3. User Research.....</b>                               | <b>4</b>  |
| <b>4. Data Collection, Processing and Preparation.....</b> | <b>5</b>  |
| <b>5. Exploratory Data Analysis.....</b>                   | <b>5</b>  |
| <b>6. Web Application Development.....</b>                 | <b>7</b>  |
| <b>7. Weekly Logbook (Weekly Notes).....</b>               | <b>8</b>  |
| <b>8. Weekly Classes Attended.....</b>                     | <b>15</b> |
| <b>9. References.....</b>                                  | <b>17</b> |

## 1. Document Version Control

| Date       | Version | Changes                           | Author           |
|------------|---------|-----------------------------------|------------------|
| 08/30/2025 | V1      | Initial template and Introduction | Ana Clara Zaidan |
| 10/07/2025 | V05     | Weekly logbooks                   | Ana Clara Zaidan |
| 10/08/2025 | V06     | Artifacts summary                 | Ana Clara Zaidan |

## 2. Introduction

This document consolidates the progress and artifacts developed for the Project titled "Intelligent Platform for Social Engagement". The project partner is the social organization "Olhar de Bia", whose mission is to end poverty through education. Ms. Beatriz Martins, the NGO's founder, acts as the Product Owner and focal point.

The main problem addressed is the difficulty in effectively connecting individuals willing to volunteer with social organizations needing collaboration. Many people give up volunteering because they don't know how to find projects or are unaware of existing institutions. Conversely, Civil Society Organizations (CSOs) face challenges in publicizing opportunities and selecting suitable volunteers, alongside difficulties in managing volunteer work. Research conducted in São Paulo (CHANG, 2022) indicated that 48.6% of respondents had never participated in volunteer work, citing reasons such as difficulty finding projects and lack of knowledge about institutions.

Given this scenario, the main objective of the project is to develop a web platform that centralizes volunteer opportunities and donation campaigns, using Artificial Intelligence to connect volunteers with causes and events best aligned with their profiles and interests.

For partner organizations, like "Olhar de Bia," the platform aims to:

- Increase the reach of potential volunteers and donors.

- Improve the "fit" between volunteers and the organization's needs/causes.

From a social impact perspective, the main objectives are:

- Stimulate participation in diverse social causes.
- Indirectly contribute to guiding resources - such as time and donations - toward organizations, thereby generating social impact.

The project started on February 4th, 2025, and is scheduled for completion on December 19th, 2025. The project is divided into four ten-week modules, and this report outlines the progress made during the third module, including **user research, data collection, processing and preparation, exploratory data analysis, environment setup, and web application development.**

### 3. User Research

A user research phase was conducted through an online questionnaire (Google Forms) aimed at understanding the demographic and behavioral profiles of potential volunteers, as well as their motivations, logistical preferences, and perceived challenges in finding suitable volunteering opportunities. A total of **42 responses** were collected.

The form included questions across six main categories:

- **Demographic data:** age, gender, and geographic location.
- **Barriers to engagement:** perceived difficulty in finding volunteer opportunities.
- **Value alignment:** perceived importance of shared values between volunteers and organizations.
- **Areas of interest:** both in terms of social causes (e.g., education, inclusion, environment) and personal development (e.g., mentoring, event organization).

- **Logistical preferences:** availability, preferred mode (in-person or remote), ideal frequency, duration, and willingness to travel.

The primary objective of this research was to train a collaborative filtering recommendation model, capable of connecting volunteers to opportunities aligned with their interests, values, and availability. The data collection targeted potential volunteers, as a first step toward building the user side of the matching system.

Basic descriptive analyses and visualizations were produced to identify initial trends and guide data modeling, and will be shown in the section “5. Exploratory Data Analysis”.

## 4. Data Collection, Processing and Preparation

All responses were exported from Google Forms and stored in a CSV file for reproducibility and analysis. Since the form design enforced mandatory fields for most questions, data quality was high, with minimal missing values — primarily limited to optional fields such as postal code and detailed time availability.

Data preprocessing included verifying response consistency, handling optional missing values, and structuring variables into categorical and numerical formats for subsequent modeling. The dataset is being prepared as the foundation for training a collaborative filtering (CF) recommendation model, which will later incorporate re-ranking mechanisms based on logistical constraints (e.g., distance, modality, time availability).

## 5. Exploratory Data Analysis

An **Exploratory Data Analysis (EDA)** was carried out using Python, leveraging libraries such as **pandas**, **NumPy**, and **Matplotlib**. The analysis aimed to identify patterns in volunteer profiles, preferences, and behaviors that could enhance the interpretability and future performance of the recommendation model.

The dataset included 37 variables describing user demographics, interests, motivations, and logistical characteristics. Key findings include:

- **Difficulty finding opportunities:** average rating of 3.3 (on a 1–5 scale), indicating moderate friction in discovering suitable events.
- **Value alignment:** mean score of 4.8, suggesting strong preference for alignment between personal and institutional values.
- **Cause interests:** top-rated areas included *social inclusion, child and youth support, education, and environment*.
- **Preferred development areas:** *mentoring/teaching* and *public engagement* were the most desired.
- **Logistics:** majority favored *in-person* participation, within *10–20 km* radius, lasting *3–5 hours*, and preferably on *weekends*.
- **Certification:** most respondents were indifferent, though a relevant subgroup valued receiving certificates.

These insights supported the construction of two key data artifacts:

- *user\_item\_preferences.csv*: a long-form dataset capturing each user's rated affinities with causes and development areas;
- *user\_profiles.csv*: a structured dataset containing availability and logistical attributes for filtering and re-ranking.

Also, the EDA results brought insights such as:

- Causes and development areas define the **item space** for collaborative filtering.
- Logistic and contextual features are applied as **re-ranking filters**.
- Value alignment is treated as a **high-weight feature** for matching relevance.
- Cold-start users will initialize their profiles through a simplified version of the same questionnaire.

## 6. Web Application Development

During this module, the **static frontend of the web platform** was developed following the interface design from Figma. The implementation was carried out in **React with TypeScript**, using **SASS** for styling. The platform includes **user authentication and session management**, integrated with **Supabase** for backend services and **Google OAuth** as the authentication provider.

The current version provides navigation across core pages (home, login, volunteer profile, events and “about us” page) and establishes the visual identity and structural foundation for future backend integration. The next stage will connect this frontend to the recommendation engine and the database, enabling dynamic content rendering and personalized volunteer event matching.

## 7. Weekly Logbook (Weekly Notes)

Designed to systematically record the project's weekly progress, the Weekly Logbook highlights the activities performed, documents challenges encountered, details solutions applied, and reflects on critical learnings. Additionally, it serves to outline planned next steps, ensuring continuous improvement and alignment with project goals.

### Week 1 (08/04/2025 to 08/10/25):

#### Activities Performed:

- Held the initial meeting with the advisor on August 5th.
- Reviewed and updated the Project Plan, incorporating refinements to the module's objectives and deliverables.
- Redefined the focus areas for Module 3, emphasizing user research, data analysis, and frontend development.
- Aligned expectations with the advisor regarding the artifacts to be produced and evaluation criteria.
- Finished the initial Environment setup with dev containers.

#### Challenges Encountered:

- Aligning project ideas with the course requirements and structure.
- Adjusting the scope to ensure a balanced workload between data-related and frontend activities.

#### Solutions Applied:

- Redefined module milestones, prioritizing completion of the frontend structure and data artifacts as foundations for the upcoming recommender system.

#### Learnings:

- Learned the importance of continuously refining the project plan to maintain alignment with the evolving scope and technical requirements..

#### Next Steps:



- Begin the new frontend implementation using React, replacing the previous Angular setup.

## **Week 2 (08/11/2025 to 08/17/2025):**

### **Activities Performed:**

- Held a meeting with the advisor on August 12th
- Started developing the frontend of the application using React (replacing Angular).
- Implemented the home page, including layout and navigation with the project's visual identity defined in Figma, with some adjustments in the colour palette, for example.

### **Challenges encountered:**

- Thinking if React was actually the best framework to choose with the available resources - such as AI support.

### **Solutions Applied:**

- Followed official React documentation to accelerate learning, and used Lovable AI to help with the development.
- Refactored the project folder structure to align with React.

### **Learnings:**

- The project is not immutable and we can adapt it to our needs as we go along.
- Gained a clearer understanding of React's component-based architecture and its advantages for scalability and maintainability.

### **Next steps:**

- Continue implementing the frontend pages, focusing on the events module and the backend integrations.

### **Week 3 (08/18/2025 to 08/24/2025):**

#### **Activities Performed:**

- Held a meeting with the advisor on August 19th
- Continued the frontend development, implementing the login page with preliminary interface components and routing.
- Integrated authentication placeholders and initial form validation logic.

#### **Challenges encountered:**

- Balancing the ambition of developing the interface all by myself manually with the realistic limitations of time.

#### **Solutions applied:**

- Utilized AI to help with the development of the components, and utilized libraries with ready to use components

#### **Learnings:**

- Understood the importance of a modular architecture for scaling UI components.
- Improved understanding of routing and event propagation in React.

#### **Next steps:**

- Finish the frontend static development

### **Week 4 (08/25/2025 to 08/31/2025):**

#### **Activities Performed:**

- Held a meeting with the advisor on August 26th.
- Developed the user profile page, integrating it with navigation and authentication placeholders.
- Explored the Supabase platform to define how user authentication and database management would be implemented in later stages.
- Set up a test Supabase project, connected it to the frontend, and configured the environment variables needed.

- Tested basic CRUD operations and authentication endpoints to validate integration.

#### **Challenges Encountered:**

- Understanding how supabase works behind the scenes and how it interacts with the react components

#### **Solutions Applied:**

- Researched and read supabase oficial documentation.

#### **Learnings:**

- Gained an understanding of Supabase as a Backend-as-a-Service (BaaS) platform and its components (PostgreSQL database, authentication, storage, and API).

#### **Next Steps:**

- Discuss if I should use supabase for my entire Database.
- Start thinking about the data collection module.

### **Week 5 (09/01/2025 to 09/07/2025):**

#### **Activities Performed:**

- Held a meeting with the advisor on September 2nd.
- Researched best practices for designing volunteer profiling surveys.
- Developed a Google Forms questionnaire including demographic, motivational, and logistical questions.
- Launched the survey and began collecting responses (42 total participants).

#### **Challenges Encountered:**

- Ensuring that the form covered all relevant aspects for both behavioral analysis and AI model training.

#### **Solutions Applied:**

- Structured the form into thematic sections and used required fields to prevent missing data.

### **Learnings:**

- Acquired practical knowledge on transforming qualitative user needs into quantifiable variables suitable for recommendation modeling.

### **Next Steps:**

- Continue data collection and proceed with exploratory data analysis (EDA).

### **Week 6 (09/08/2025 to 09/14/2025):**

#### **Activities Performed:**

- Advanced the frontend implementation, refining layout, responsive design, and user navigation.
- Prepared the project environment for data integration in the next stages.

#### **Challenges Encountered:**

- Balancing frontend progress with the upcoming data analysis workload.

#### **Solutions Applied:**

- Allocated fixed time blocks for each workstream to ensure balanced progress.

### **Learnings:**

- Improved personal time management when handling parallel tasks.

### **Next Steps:**

- Begin data analysis using Python libraries.

### **Week 7 (09/15/2025 to 09/21/2025):**

#### **Activities Performed:**

- Held a meeting with the advisor on September 16th.
- Conducted the Exploratory Data Analysis (EDA) of the Google Forms responses.
- Generated descriptive statistics and visualizations for each question, identifying user patterns and preferences.

### **Challenges Encountered:**

- Dealing with multiple variable types (scales and multi-select questions) in the same dataset.

### **Solutions Applied:**

- Used libraries such as pandas, NumPy, and Matplotlib.

### **Learnings:**

- Learned how to derive insights from small datasets and prepare them for modeling and future filtering operations.

### **Next Steps:**

- Document the analysis results and synthesize insights relevant to the recommender design.

## **Week 8 (09/22/2025 to 09/28/2025):**

### **Activities Performed:**

- Held a meeting with the advisor on September 23rd.
- Documented the EDA findings and insights, compiling them into a structured markdown report with figures and interpretations.
- Outlined how these insights would guide the development of the collaborative filtering recommender and future re-ranking logic.

### **Challenges Encountered:**

- Summarizing quantitative insights into actionable design implications for the recommender system.

### **Solutions Applied:**

- Created clear visual explanations (“What it means,” “How to read,” and “How to interpret”) for each chart to improve readability.

### **Learnings:**

- Translating analytical outcomes into practical recommendations with a small dataset.

### **Next Steps:**

- Discuss about the model itself.

## **Week 9 (09/29/2025 to 10/05/2025):**

### **Activities Performed:**

- Held a meeting with the advisor on September 30th.
- Connected the login page with the Supabase backend to enable real session handling.
- Validated the authentication flow end-to-end.

### **Challenges Encountered:**

- Thinking if supabase should be directly connected to the front end

### **Solutions Applied:**

- Followed official Supabase documentation and discussed with the advisor about rethinking this part of the solution architecture.

### **Learnings:**

- Improved decision making regarding deliverable prioritization.
- Gained practical experience with backend-as-a-service platform

### **Next Steps:**

- Finalize module documentation and prepare the final presentation.

## **Week 10 (10/06/2025 to 10/09/2025):**

### **Activities Performed:**

- Held a meeting with the advisor on October 8th.
- Consolidated all artifacts produced during the module, including datasets, analyses, and frontend components.
- Prepared the final presentation and module public report, summarizing technical progress and insights.

### **Challenges Encountered:**

- Encountered challenges in task prioritization.
- Time management between report writing, presentation design, and final artifact refinement.

### **Solutions Applied:**

- Utilized one demand to help develop another. Example: utilized content used in the report also in the final presentation.
- Focused efforts on completing deliverables from the current sprint, while deferring tasks or revisions from previous sprints to a later stage.

#### **Learnings:**

- Understood the importance of integrating documentation continuously throughout the development cycle rather than only at the end.

#### **Next Steps:**

- Prepare for the next module by initiating integration between the frontend and backend
- Developing the recommendation system.

## **8. Weekly Classes Attended**

This section lists the mandatory classes attended every Tuesday during the analyzed period.

- **August 5, 2025:** CAREER TRACK – Graduate Studies Preparation: Differences Between Programs
- **August 12, 2025:** CAREER TRACK – Graduate Studies Preparation: Application Processes in Brazil
- **August 19, 2025:** CORPORATE TRACK – Project Management and Methodologies: Project Management with Artificial Intelligence
- **August 26, 2025:** ENTREPRENEURIAL TRACK – Product Design: Key Components of a Business Plan
- **September 2, 2025:** CORPORATE TRACK – Labor Market in Computing and Certifications: Adapting to Technological Changes
- **September 9, 2025:** ENTREPRENEURIAL TRACK – Strategic People Management and Culture: Leadership and Organizational Culture
- **September 16, 2025:** CAREER TRACK – Graduate Studies Preparation: Application Processes Abroad

- **September 23, 2025:** CAREER TRACK – Graduate Studies Preparation: Decision-Making Process
- **September 30, 2025:** CORPORATE TRACK – Corporate Venture Capital and M&A: Post-Acquisition Integration Strategies
- **October 7, 2025:** CORPORATE TRACK – Intellectual Property and Patents: International Protection of Intellectual Property



## 9. References

CHANG, Fernanda Li. **Match voluntário: conexão entre organizações da sociedade civil e voluntários**. 2022. Trabalho de Conclusão de Curso (Bacharelado em [...] – Universidade [...], Limeira, 2022. Nota: Details like degree, course, and university for Chang's work were not in the provided text and need to be filled in.